(6) Suppose that f: $\mathbb{R}^2 \to \mathbb{R}$ is differentiable everywhere and that $y = f(x_1, x_2)$ where

$$x_1 = r\cos\theta$$

$$x_2 = r\sin\theta.$$

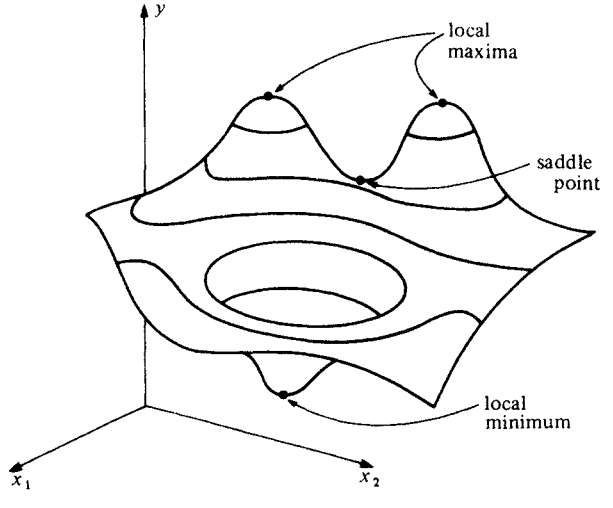
Express $\partial y/\partial x_2$ in terms of $\partial y/\partial r$ and $\partial y/\partial\theta$.

19.27 Stationary points

In §11.3 we defined a point ξ to be a *stationary point* of f if $f'(\xi) = 0$. We shall use precisely the same definition in the case of a real-valued function of a vector variable. In this case, of course, $f'(\xi)$ is a $1 \times n$ row vector and $f'(\xi) = 0$ means that all the co-ordinates of this row vector (i.e. the partial derivatives of f at ξ) are zero. In geometric terms, the fact that ξ is a stationary point means that the tangent hyperplane

$$y = f(\xi) + f'(\xi)(x - \xi)$$

reduces to $y = f(\xi)$, i.e. the tangent hyperplane is horizontal at a stationary point. The diagram shows some points on a surface $y = f(x_1, x_2)$ at which the tangent plane is horizontal.



Any local maximum or minimum at which f is differentiable is, of course, a stationary point. This follows immediately from corollary 19.9 and we quote the result as a theorem below. It is not at all uncommon, however, for stationary points in the vector case to correspond to *saddle points* as indicated in the

diagram. Less common, but not to be ignored, are other more exotic stationary points similar to those mentioned in §11.3.

19.28 Theorem Suppose that $f: A \to \mathbb{R}$ where A is a set in $\mathbb{R}^n$. If f is differentiable at $\xi \in A$ and f has a local maximum or minimum at ξ, then

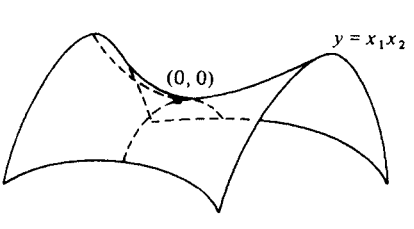
$$f'(\xi) = 0.$$

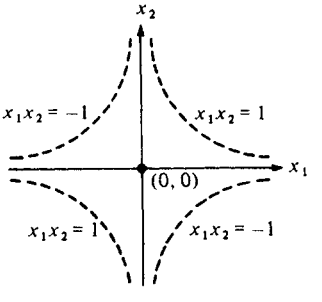
Proof This follows from corollary 19.9 and theorem 19.16.

19.29 Example Consider the function $f: \mathbb{R}^2 \to \mathbb{R}$ defined by $f(x_1, x_2) = x_1x_2$. Observe that

$$f'(0, 0) = (f_1(0, 0), f_2(0, 0)) = (0, 0)$$

but f does *not* have a local maximum or minimum at (0, 0). On the line $x_1 = x_2$ we have that $f(x_1, x_2) = x_1^2 > 0$ unless $x_1 = 0$ while, on the line $x_1 = -x_2$, $f(x_1, x_2) = -x_1^2 < 0$ unless $x_1 = 0$.





The two diagrams are alternative representations of the situation. The left hand diagram makes it clear why f is said to have a saddle point at (0, 0).

19.30 Example In 'least squares analysis', one is given an *inconsistent* system $Ax = b$ of linear equations and asked to find the vector or vectors x which minimises $\|Ax - b\|$. Consider

$$y = \|Ax - b\|^2 = (Ax - b)^{\mathrm{T}}(Ax - b) = u^{\mathrm{T}}u$$

where $u = Ax - b$. By formula (V) of §19.21,

$$\frac{dy}{dx} = \frac{dy}{du}\frac{du}{dx} = (2u^{\mathrm{T}}I)A = 2(Ax - b)^{\mathrm{T}}A.$$

At a local minimum, the derivative must be the zero vector and so, if x minimises $\|Ax - b\|$, then